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Influence of oxygen vacancies and electrical properties in $Ba_{0.92}Ca_{0.08}Zr_{0.05}Ti_{0.95}O_3$ ceramics through the co-doping of La^{3+} and Tm^{3+} ions

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ABSTRACT

Barium calcium zirconate titanate (BCZT)-based piezoelectric ceramics, which are modified with rare earth elements, are one of the candidates for replacing lead-based piezoelectric ceramics. In this study, the BCZT-0.8 % La-*x*Tm ceramics were prepared using the sol-gel method. The effects of varying Tm^{3+} ions contents on the dielectric, ferroelectric, and piezoelectric properties of ceramics were investigated and the influence mechanism was discussed. Results showed that La^{3+} and Tm^{3+} ions co-doping can affect the motion of domains and domain walls by adjusting oxygen vacancy concentrations and microstructure, which is the main reason for the various electrical properties of ceramics. Especially when the Tm^{3+} ions content is 2 %, many oxygen vacancies would generate, which is due to Tm^{3+} ions replace to B-site cations. These oxygen vacancies would produce pinning effect for the domains, which would reduce the electrical properties of the ceramics. When Tm^{3+} ions content was 0.7 %, the ceramics showed excellent dielectric constant ($\varepsilon_r = 9941.6$), remnant polarisation ($P_r = 28.93$ μC/cm), and piezoelectric constant ($d_{33} = 181$ pC/N). The results have some reference value for enhancing the electrical properties of BCZT piezoelectric ceramics using elements co-doping.

1. Introduction

Piezoelectric ceramics are significant novel functional materials commonly utilised in devices such as sensors, pulsed power devices, and transducers [1,2], playing a crucial role in the development of modern industry. Lead-based piezoelectric ceramics are currently the most widely commercialised ceramics due to their outstanding performance, while their environmental and health hazards cannot be ignored [3,4]. Therefore, substituting lead-based piezoelectric ceramics with lead-free piezoelectric ceramics has great significance. Barium calcium zirconate titanate (BCZT)-based lead-free piezoelectric ceramics are regarded as one of the most promising candidate materials [5,6], because they could enhance electrical properties by constructing the morphotropic phase boundaries and polymorphic phase transitions [7,8]. Nonetheless, despite the relatively outstanding performance of BCZT ceramics, some defects still exist. For example, they have a low Curie temperature, and their electrical properties still cannot be compared with those of lead-based piezoelectric ceramics.

To enhance the performance of BCZT ceramics, researchers have focused on the doped elements method. In recent years, BCZT piezoelectric ceramics have obtained high Curie temperature, low dielectric loss, high remnant polarisation and high piezoelectric constant by doping different elements to regulate the internal microstructure, defects and multiphase coexistence of ceramics [9,10]. For example, Kumari et al. [11] prepared Bi^{3+} ions doped $Ba_{0.98-3x/2}Bi_xCa_{0.02}Zr_{0.02}Ti_{0.976}Cu_{0.008}O_3$ lead-free ceramics using the solid-state method. In their work, the grain size decreased significantly from 16.14 to 2.11 μm with increased Bi^{3+} ions contents. When Bi^{3+} ions content was 0.02, the ceramics had high Curie temperature (159 °C), high dielectric constant ($\varepsilon_r = 3146$), and low dielectric loss ($\tan\delta = 0.019$), meanwhile, the

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temperature stability of electrical properties for the ceramics was improved. Li et al. [12] prepared $Ba_{0.85}Ca_{0.15}Zr_{0.1}(Ti_{1-x}Ni_x)_{0.9}O_3$ (BCZT-xNi) ($x = 0$, 0.01, 0.02, 0.04) ceramics using the sol-gel method. When $x = 0.01$, the ceramics had a high dielectric constant ($\varepsilon_r = 4473$) and good frequency stability. Meanwhile, a low dielectric loss (tan $\delta = 0.004$) at 100 Hz was detected in their work. Liu et al. [13] prepared Li-doped $(Ba_{0.85}Ca_{0.15})(Zr_{0.1}Ti_{0.9})O_3$ ceramics. The presence of Li dopants reduced the sintering temperature of the ceramics, which then improved its compactness greatly. The best results in this work showed that the $\varepsilon_r = 3814$, tan $\delta = 0.01$, $d_{33} = 420$ pC/N, and $k_p = 47$ %. Zheng et al. [14] prepared $Ba_{0.85}Ca_{0.15}Zr_{0.1}(Ti_{1-x}Mn_x)_{0.9}O_3$ ($x = 0$, 2 %, 4 %, and 8 %) ceramics using the sol-gel method. Their results indicated that Mn could inhibit the growth of ceramic grain size and the Curie temperature would decrease with the increase of Mn content. When $x = 4$ %, the ceramics exhibited a high dielectric constant ($\varepsilon_r = 3021$ at 100 Hz) and a low dielectric loss (tan $\delta = 0.0569$ at 10^2 Hz). Ramovatar et al. [15] fabricated $Ba_{0.98}Ca_{0.02}Zr_{0.02}Ti_{0.98}O_3$-$x$Pr piezoelectric ceramics via a solid-state method. When Pr^{3+} ions content was 0.04, the ceramics possessed a large remnant polarisation ($2P_r$, ~ 13 $\mu C/cm^2$) and low coercive field (Ec, ~ 22 V/cm) with the co-existed tetragonal and orthorhombic phases. The results indicated that the dielectric, ferroelectric, and piezoelectric properties of BCZT ceramics were significantly improved by various ion dopants. Despite the high piezoelectric constant and low dielectric loss obtained via ceramic doping, the dielectric constant and Curie temperature of the ceramics were relatively low. Hence, enhancing the dielectric constants and performance stability near Curie temperature of the ceramics is a significant task.

In this study, piezoelectric ceramics were fabricated via the sol-gel method, and the mechanism of defect dipoles generated by altering the concentrations of doping rare earth ions was explored. The enhancement of dielectric properties and the occurrence of diffuse phase transitions in ceramics facilitate their utilization near the Curie temperature. Simultaneously, the influences of co-doping with rare earth ions on the electrical properties of the ceramics were investigated in a relatively systematic manner from aspects such as microstructure and defects.

2. Experimental procedures

2.1. Preparation

Ceramic samples of $Ba_{0.92}Ca_{0.08}Ti_{0.95}Zr_{0.05}$-0.8 %La-$x$Tm (BCZT-0.8 %La-$x$Tm, $x = 0$, 0.1 %, 0.3 %, 0.5 %, 0.7 %, 1 %, and 2 %) were synthesised using the sol-gel method. High-purity materials, including $Ba(CH_3COO)_2$ (A.R.), $Ca(NO_3)_2{\cdot}4\ H_2O$(98.5 %), $Zr(NO_3)_4{\cdot}5\ H_2O$ (A.R.), $Ti(OC_4H_9)_4$ (C.P.), $La(NO_3)_3{\cdot}nH_2O$ (A.R.), and $Tm(NO_3)_3{\cdot}5\ H_2O$ (C.P.) were used as raw precursors. The raw precursors were added to an Erlenmeyer flask, and then ethanol was added to react for 3 h to obtain polymer solutions. Subsequently, the solutions were placed in the oven at 120 °C for 10 h and then calcined at 900 °C for 2 h in a muffle furnace to form precursor powders. Subsequently, the powders were mixed with 2 wt% polyvinyl alcohol (PVA) as an organic binder (the weight ratio of PVA and powders was 1:120) were formed into disks with a diameter of 20 mm under 20 MPa pressure with a retention time of 150 s. The disks were embedded in alumina and sintered at 1300 °C for 4 h in air using a heating rate of 5 °C/min.

2.2. Characterisation and measurement

X-ray diffraction (XRD, Rigaku Smartlab, Japan) with Cu-Kα radiation was used to characterise the phase identification for precursor powders and ceramics in Bragg's angle range of 10°–80°. The sweeping rate, voltage, and current for XRD were 0.01°/min, 40 kV, and 15 mA, respectively. The concentrations of oxygen vacancy were tested with X-ray photoelectron spectroscopy (XPS, K-ALPHA, UK). The microstructure on polished surfaces for samples was studied by scanning electron microscopy (SEM, Phenom Prox, Netherlands). A dielectric testing instrument (HDMS-1000V, China) and an LCR instrument (WK-6500B, UK) were used to characterise the relative dielectric constant (ε_r) and dielectric loss (tan δ) of ceramics that coated a silver electrode in different frequencies and temperatures (30–180 °C). Meanwhile, the mechanic quality factor (Q_m) of ceramics was tested at room temperature in these instruments. The radiant precision workstation (RTI-Multiferroic II, USA) was used to obtain the polarisation-electric field (P-E) hysteresis loops under the electric field at 30 kV/cm and 2 Hz. The piezoelectric constant (d_{33}) was tested using a quasistatic piezoelectric meter (ZJ-3, China).

3. Results and discussion

3.1. Phases and structures

Fig. 1(a) and (c) display the XRD patterns of BCZT-0.8 %La-xTm precursor and ceramic powders, respectively, at 2θ = 10°–80°. Clearly, all the ceramics observed a typical perovskite structure without an impurity phase. The results indicated that the La^{3+} and Tm^{3+} ions were completely diffused into the lattice of the ceramics. At the same time, the precursor and ceramic powders showed strong and sharp XRD peaks, which proved that the ceramics had good crystalline structure. The trends shown in Fig. 1(b) and (d) for the characteristic peak (44°–46°) shifting to lower angles could be elucidated through the ionic radius. Tm^{3+} ions will occupy the B site of the ABO_3 structure due to their small ionic radius (r = 0.869 Å) [16]. The ionic radii of Ti^{4+} and Zr^{4+} (B site of ABO_3) ions were 0.68 and 0.80 Å. Therefore, the characteristic peak shifts to lower angles due to the lattice contraction when Tm^{3+} ions replaced the site of Ti^{4+} and Zr^{4+} ions. The relationship between diffraction angle and interplanar crystal spacing is shown by Bragg's law (Eq. 1).

$$2d\sin\theta = n\lambda \tag{1}$$

where d is the interplanar crystal spacing, θ is the diffraction angle, and λ is a length of wave. Meanwhile, the 1:2 split of the characteristic peak is shown in Fig. 1(b) and (d). According to the PDF card (PDF # 79–2265), this schismatic characteristic peak occurs in ceramics when the ceramics possess a tetragonal (T) phase.

To further study the influence on lattice parameters when Tm^{3+} ions enter the lattice, the XRD patterns ($x = 0$, 0.7 %, and 2 %) were refined using the GASA II program. The results are shown in Fig. 2 and Table 1. The diminutive difference plot indicates that the refined data match well with the XRD patterns. Moreover, the confidence factor (R_{wp}) ranging from 6.9 % to 7.7 % in Table 1, demonstrating the high reliability of the refined data. The Table 1 shows the rise in lattice parameters with the increase of x. The change in lattice parameters may also be related to oxygen vacancy, according to the literature [17]. In addition, the refinement results likewise confirmed that the ceramics showed a T phase, which was consistent with the analysis of XRD patterns.

The XPS results and corresponding fitting data for O1s for BCZT-0.8 %La-xTm ceramics with various x contents ($x = 0$, 0.3 %, 0.7 %, and 2 %) are showed in Fig. 3. The three peaks of the XPS spectrum from 527 to 533 eV consisted of the oxygen lattice (cation oxygen bonds), oxygen vacancy and absorbed H_2O, successively [18–20]. And the ratio of oxygen vacancies ($O_{Vacancy}/O_{Lattice}$) is exhibited in Fig. 3(e). The results indicated that the concentrations of oxygen vacancy rose with the increase of x. The main reason for this was Tm^{3+} ions replaced Ti^{4+} and Zr^{4+} ions, which led to a difference in the valence (Fig. 4). On the one hand, the incremental oxygen vacancy concentrations increase the conductivity, which led to high dielectric loss. On the other hand, a large number of V_O-$Tm'_{Ti/Zr}$ dipole defects are produced when the oxygen vacancy concentrations are high. The defect equation is shown in Eq. 2. These dipole defects would accumulate near grain boundaries and domain walls, which affected the electrical properties of the ceramics

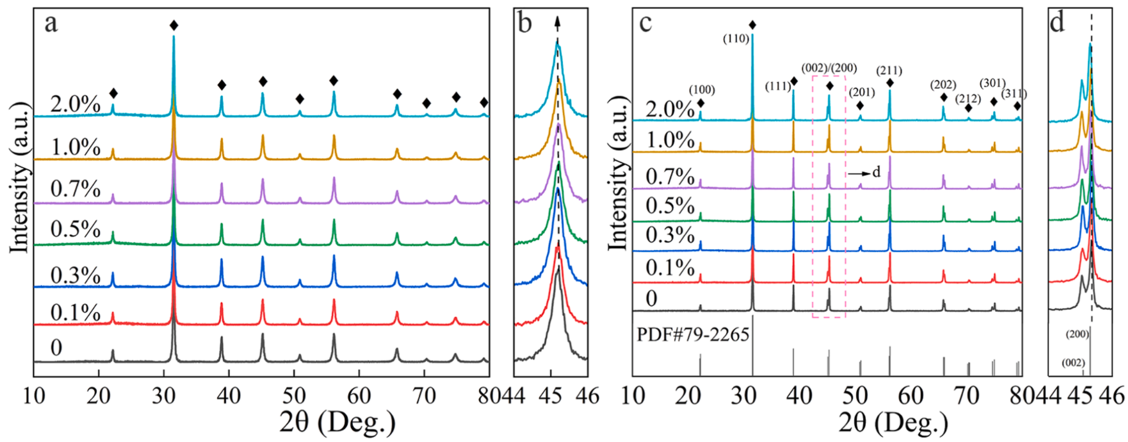

Fig. 1. (a) XRD patterns of precursor powders, (b) enlarged images of precursor powders, (c) XRD patterns of BCZT-0.8 %La-*x*Tm ceramics powders and (d) enlarged images of ceramics powders.

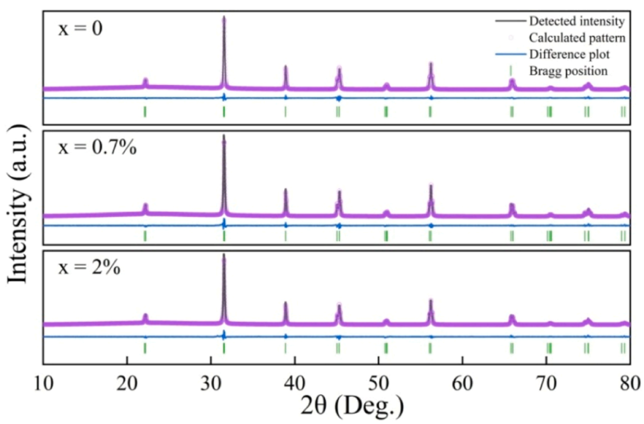

Fig. 2. XRD refinement patterns of BCZT-0.8 %La-*x*Tm ceramics powders (x = 0, 0.7 %, and 2 %).

Table 1
Refinement results of the BCZT-0.8 %La-*x*Tm ceramics.

x	R_{wp}	a	c
0	6.9	3.9940	4.014
0.7 %	7.1	3.9946	4.018
2 %	7.7	3.9961	4.017

through inhibiting grain growth, domain motion, and the switch of domain walls [21–23].

$$2Tm^{3+}+O_O^{\times}\xrightarrow{BCZT-La-xTm}2Tm'_{Ti/Zr}+V_O+\frac{1}{2}O_2\uparrow \tag{2}$$

Fig. 5 reveals the SEM images of the polished surfaces of BCZT-0.8 % La-*x*Tm ceramics. The ceramics have a dense microstructure. And their grain size appeared to increase initially and then decrease when x changed from 0 % to 2 %. This phenomenon was attributed to the incremental concentrations of oxygen vacancy [24]. The mass transfer appeared to an increasing trend when the concentrations of oxygen vacancy rose, which would promote the growth of grain size. However, with higher concentrations of oxygen vacancy than before in the ceramic structure, the grain size was reduced. In addition, the lattice distortion inhibited the grain boundary migration when Tm^{3+} ions occupied the B sites, which was also an important reason for the reduction in grain size [20].

3.2. Dielectric properties

Fig. 6(a) shows the temperature-dependent relative permittivity and dielectric loss of BCZT-0.8 %La-*x*Tm ceramics with different x contents at a frequency of 10 kHz. The relative permittivity trended to increase and then decrease with the increase of x contents. The reason for this effect was a mass of charges accumulated at large grain boundaries as grain size continued to grow, which would generate a strong potential barrier [25]. This outcome could cause high relative permittivity for ceramics. However, the grain size growth and motion of domains were inhibited with the increase of oxygen vacancy concentrations [26]. This would lead to a reduction in the dielectric constant. In addition, the strong potential barrier at large grain boundary made a low dielectric loss for ceramics [27], but the dielectric loss showed a trend of rising near the Curie temperature (Tc), which was due to the rapid increment of conductivity at a high temperature [28]. In addition, the change of carrier concentrations by ions doping was the other reason for influencing the leakage current of the ceramics [29]. The dielectric peak corresponds to the Tc, which signifies a transformation from the paraelectric phase to the ferroelectric phase. Meanwhile, the Tc values decreased gradually with increasing x. The specific Tc values are displayed in Fig. 6(b). The long-range order of ceramics was destroyed when Tm ions were doped in ceramics. This result would cause high chemical disorder and local heterogeneity, leading to lower Tc and more diffuse phase transitions than before [30]. In addition, the maximum permittivity of ceramics decreased gradually with an increment of frequencies (Table 2), which was attributed to the synergistic effects of diverse polarization mechanisms [31,32].

Fig. 7 shows the temperature dependence of inverse permittivity ($10^4/\varepsilon_r$) of BCZT-0.8 %La-*x*Tm ceramics at a frequency of 10 kHz, the temperature deviation and the relationship of ln (T - T_m) assessed with ln ($1/\varepsilon_r$ - $1/\varepsilon_m$) of BCZT-0.8 %La-*x*Tm ceramics with various x contents. The Curie-Weiss law and modified Curie-Weiss law are shown in Eqs.3–5.

$$\frac{1}{\varepsilon_r}=\frac{T-T_{cw}}{C}(T>T_{cw}) \tag{3}$$

$$\Delta T_m=T_B-T_m \tag{4}$$

$$\frac{1}{\varepsilon_r}-\frac{1}{\varepsilon_m}=\frac{(T-T_m)^{\gamma}}{C'} \tag{5}$$

where C expresses the Curie-Weiss constant, T_{cw} represents the Curie-Weiss temperature, ΔT_m corresponds to the temperature deviation of Curie-Weiss law (dielectric diffusion degree), T_B is the temperature which the dielectric constant begins to obey Curie-Weiss law, T_m is the

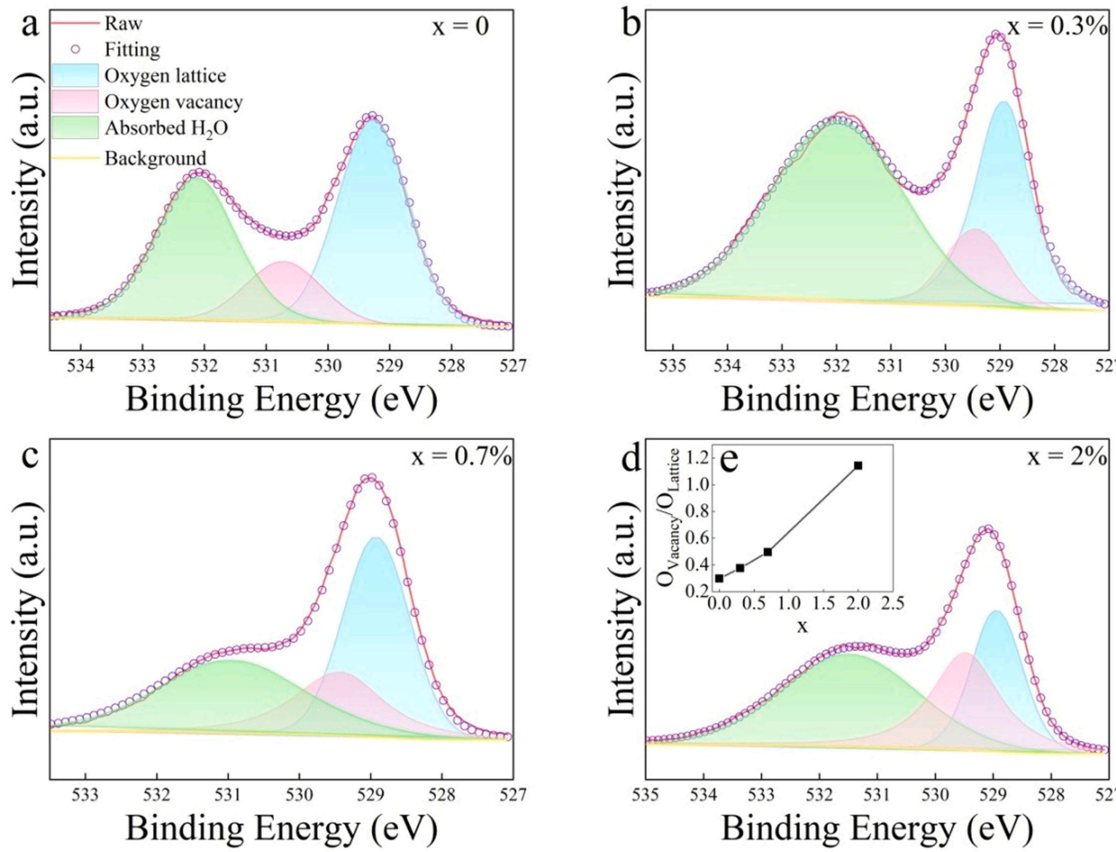

Fig. 3. XPS spectrum of O1s for BCZT-0.8 %La-xTm ceramics: (a) $x = 0$, (b) $x = 0.3$ %, (c) $x = 0.7$ %, (d) $x = 2$ %, and (e) the ratio of oxygen vacancies ($O_{Vacancy}/O_{Lattice}$).

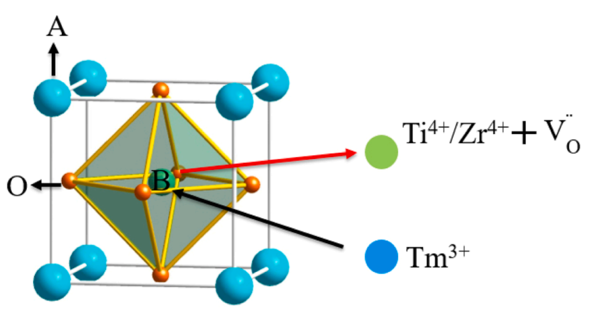

Fig. 4. Simple diagram of Tm ions replacing the B site for ABO_3.

temperature corresponding to the dielectric constant, C' is the modified Curie-Weiss parameter, and γ corresponds to the constant of diffuseness via the relationship of ln (T - T_m) assessed with ln ($1/\varepsilon_r$ - $1/\varepsilon_m$). When the value of γ is 1, the samples show a normal ferroelectricity. With the incremental γ value at 2, the samples transform to a relaxor ferroelectricity. Slightly reduced ΔT_m and γ values were found when x contents changed from 0 % to 0.1 %. The result indicated that the small thulium dopant could reduce the diffuseness phase transformation of ceramics. With the x contents continuing to rise, the γ values increased gradually to 1.4685 from 1.1950, and the ΔT_m showed an increasing trend. This phenomenon was attributed to the increased disorder and volatility of the local composition [33,34].

3.3. Ferroelectricity properties

Fig. 8 illustrates the *P-E* loops for BCZT-0.8 %La-xTm ceramics. The remnant polarisation (P_r) and coercive field (Ec) initially increased gradually with the increase of x contents, and the maximum P_r was equal to 28.9 μC/cm^2 when x was 0.7 %. This outcome was attributed to the domains easily switched in a large grain size [35], and the distortion of oxygen octahedron when Tm^{3+} ions entered crystal structure [36]. Meanwhile, the diffusion behaviour of the ceramics deteriorated with the growth of grain size, which led to high P_r and Ec of the ceramics [37]. The ceramics appeared T phase was another reason for the rise of Ec [38]. However, with the further increase of x contents, the oxygen vacancy concentrations also rose. This result could promote the generation of V_O-$Tm'_{Ti/Zr}$ dipole defects, thus pinning the ferroelectric domains and causing increase carrier migration, which led to the P_r for ceramic reduction [39]. The results were consistent with the conclusion obtained via XPS spectrum analysis. The perturbation of ferroelectric order with incremental x was also a reason for the decrease of P_r [40]. The thinned *P-E* loops and decreased Ec with large x contents could be attributed to the presence of polar nanoregions (PNRs) and the disruption of ferroelectric long-range ordering [40–42].

3.4. Piezoelectric properties

The piezoelectric constant (d_{33}) and mechanical quality factor (Q_m) of BCZT-0.8 %La-xTm ceramics are shown in Fig. 9. The d_{33} increased first and then decreased with the increment of x contents. Furthermore, the d_{33} reached peak value when x contents were equal to 1 %. This result was due to the motion of the domain walls, and the domains being switch easily in large grain sizes [43,44]. The distortion of the ceramics' oxygen octahedron structure was also a reason for the optimization of d_{33} [45]. The Q_m of samples appeared to have an opposite trend to the variation tendency of d_{33} values, and the maximum Q_m value was shown when x was equal to 2 %. Meanwhile, the d_{33} displayed a decreasing trend when x contents increased from 1 % to 2 %. The phenomenon was

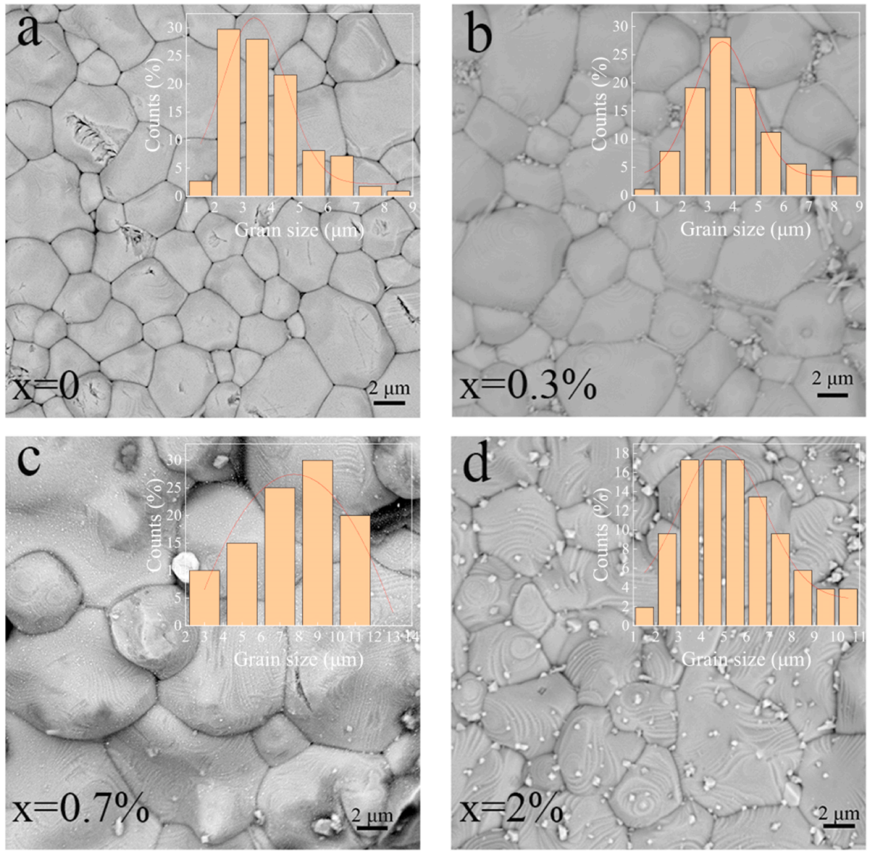

Fig. 5. SEM images and grain-size distribution of BCZT-0.8 %La-xTm ceramics: (a) x = 0, (b) x = 0.3 %, (c) x = 0.7, and (d) x = 2 %.

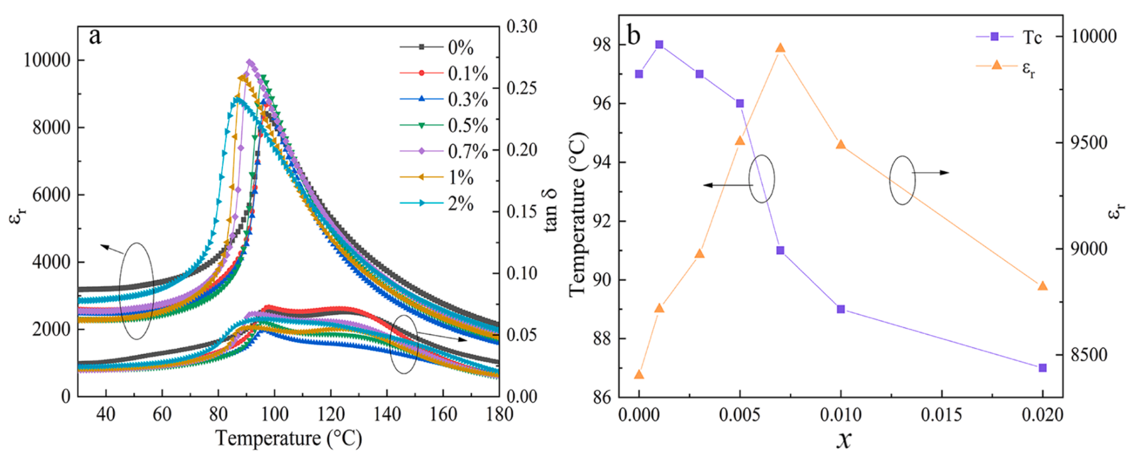

Fig. 6. (a) Temperature dependence of dielectric constant (ε_r) and dielectric loss (tan δ) of BCZT-0.8 %La-xTm ceramics with various x contents at a frequency of 10 kHz, (b) ε_{max} and Tc of BCZT-0.8 %La-xTm ceramics with different x contents.

Table 2
The maximum relative permittivity of BCZT-0.8 %La-0.7 %Tm ceramic at varying frequencies.

Frequency (Hz)	100	1k	10k	100k	1 M
ε_{max}	13853.8	11143.2	9941.6	9170.6	8575.9

due to the incremental concentrations of oxygen vacancy pinning the domains and domain walls, which led to the increase of Q_m and the decrease of d_{33} [46–48]. This finding was basically consistent with the results of XPS spectrum analysis. In addition, the dense microstructure, reduction of strain energy and multiphase coexistence could improve the piezoelectric properties of ceramics [49–51].

4. Conclusions

BCZT-0.8 %La-xTm co-doped ceramics were prepared using the sol-gel method in this study. The results indicated that the La^{3+} and Tm^{3+} ions could bring about high dielectric, ferroelectric, and piezoelectric properties for ceramics by adjusting the oxygen vacancy concentrations. Doping the La^{3+} and Tm^{3+} ions in the ceramics would also achieve a dense microstructure for the ceramics, thus improving the electrical properties of the samples. And, an appropriate amount of Tm^{3+} ions

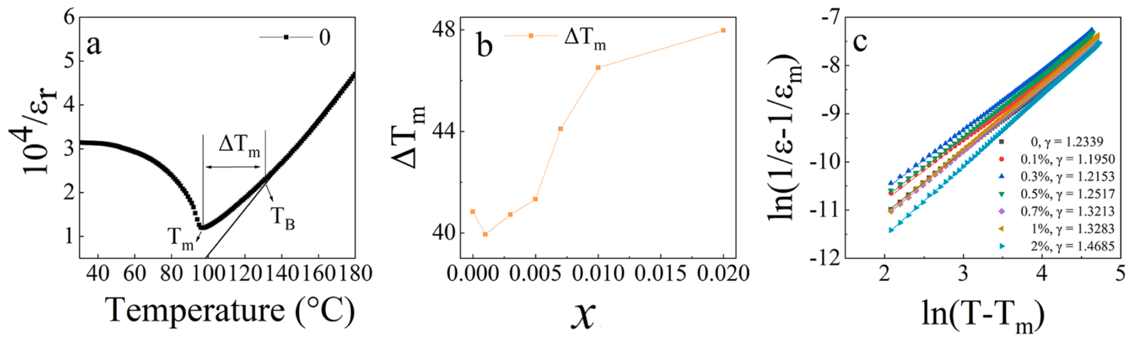

Fig. 7. (a) Inverse permittivity ($10^4/\varepsilon_r$) of BCZT-0.8 %La-xTm ceramics, (b) temperature deviation (ΔT_m) of BCZT-0.8 %La-xTm ceramics with various x contents, and (c) chart of ln ($1/\varepsilon_r$ - $1/\varepsilon_m$) vs ln (T - T_m) of the BCZT-0.8 %La-xTm ceramics with different x contents.

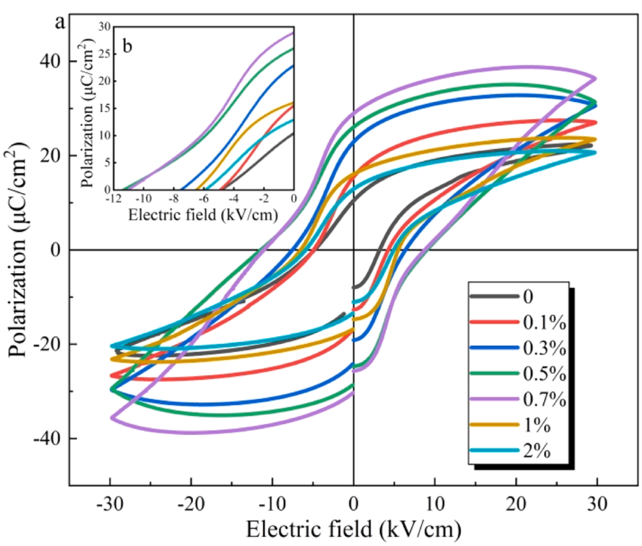

Fig. 8. (a) Ferroelectric P-E loops of BCZT-0.8 %La-xTm ceramics with various x contents, (b) enlarged images of P-E loops.

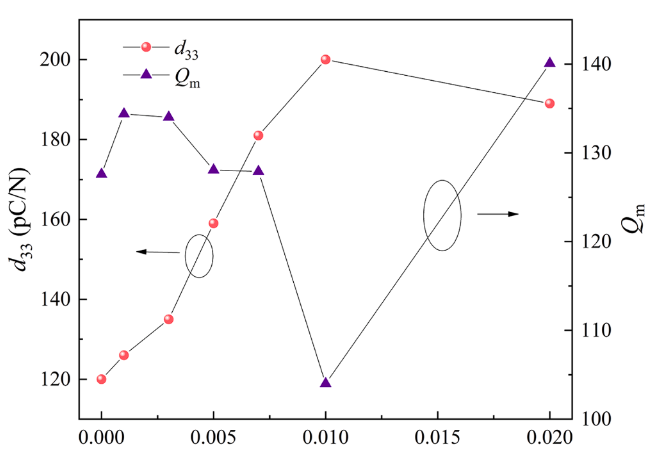

Fig. 9. Piezoelectric constant (d_{33}) and mechanical quality factor (Q_m) of BCZT-0.8 %La-xTm ceramics with various x contents.

dopant could increase the grain size of ceramics. At the same time, the disorder and volatility of the local composition increased along with the rise of Tm^{3+} ions contents, which led to the diffuse phase transitions of ceramics. This result could result in ceramics with better applications to various devices. When Tm^{3+} ions content was 0.7 %, the ceramics showed the excellent dielectric constant (9941.6), maximum remnant polarization (28.9 μC/cm), and piezoelectric constant (181 pC/N). And meaningful T_C and Q_m of the ceramic was 93 °C and 127.9, respectively. In summary, co-doping with rare earth ions can bring about excellent electrical properties for BCZT-based ceramics, making them a potential lead-free piezoelectric candidate.

CRediT authorship contribution statement

Qiangshan Jing: Supervision, Conceptualization. **Jinshuang Wang:** Validation, Software, Funding acquisition. **Xiongjie Hu:** Methodology, Conceptualization. **Xiang Ji:** Validation, Conceptualization. **Dongxu Sheng:** Investigation, Data curation. **Chunying Liu:** Validation, Software, Investigation, Data curation. **Mingyang Ma:** Writing – review & editing, Software, Methodology, Investigation, Data curation. **Yongshang Tian:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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